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Efficiency vs. Exponential Demand: Why the Natural Limit of AI Is Power

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“AI is a learning machine. And in network-effect businesses, when the learning machine learns faster, everything accelerates. It accelerates to its natural limit. The natural limit is electricity. Not chips, electricity, really.”— Eric Schmidt

Let’s start this paper with a Mike Tyson uppercut. **Efficiency** is the most overused word in the AI debate. Ok, I said it. For those fading AI, efficiency is hope. For those riding AI, efficiency is fear. Efficiency cannot be thought of in a vacuum. Embedded in the word is a hidden assumption about the pace of demand. Which brings us back to this quote: *“Jevons paradox strikes again! As AI gets more efficient and accessible, we will see its use skyrocket, turning it into a commodity we just can’t get enough of.”* That was Satya Nadella on X after DeepSeek. He was 100% correct then, but people keep coming back to efficiency as if it were a magic bullet solution just days away. Efficiency gains are like buying bigger clothes for a baby, you feel caught up for a moment, but the baby keeps growing so fast you’re instantly behind again. In AI, efficiency is the hope, but exponential demand stats are the facts, and the data already shows that growth in reasoning, tokens, and power needs is outpacing every gain. It is a certainty that there will absolutely be continuous efficiency improvements along the way, but they won’t change the bigger truth: the massive wave of AI demand acceleration is still in front of us, and power is **now** the real story.

On a recent Moonshots with Peter Diamandis podcast, the panel discussed in detail the release of ChatGPT-5 and its importance. This podcast came out just before the latest “AI bubble” warnings and fears hit the market. What made this release different was that GPT-5 introduced a router that automatically directs prompts to the right model in the background. Users no longer have to choose; reasoning is simply built in. That design choice turned reasoning from an advanced option into the default experience. On the podcast, they said the release of GPT-5 marked a turning point in the democratization of reasoning, with *“700 million people, many of them free-tier users, now using reasoning for the first time.”*

Unlike DeepSeek, which gave the world an open-source reasoning model but was mostly adopted by those already advanced in AI, GPT-5 removes that friction entirely. Now, everyday users *“who don’t know which model to use”* are benefiting from the same powerful tools. And reasoning is not a lightweight feature. Unlike simple Q&A, it triggers multi-step chains of thought, extended context use, and higher token consumption. Each reasoning call is orders of magnitude more compute-intensive than a casual prompt. When democratized across hundreds of millions of users, that translates directly into exponential growth in tokens and power consumed. As one host put it, this shift *“starts to recurse back through the system and feed more real capital into the system to build more data centers and empower more people.”* Revolutionary capability equals mass demand, and that demand is increasingly constrained not by model availability, but by the electricity required to sustain it. Efficiency will matter, but history shows it will always be outrun by adoption.

Since the launch of GPT-5, Sam Altman has publicly confirmed a dramatic surge in reasoning usage, noting that it jumped from <1% to 7% for free users, and from 7% to 24% for Plus users. OpenAI responded by increasing rate limits for reasoning calls and promising that *“all model-class limits will shortly be higher than they were before GPT-5.”* Altman framed this as necessary to keep up with demand, saying, *“I expect use of reasoning to greatly increase over time, so rate limit increases are important.”* He also acknowledged a serious capacity crunch *“We have better models, and we just can’t offer them because we don’t have the capacity”* and forecast *“trillions of dollars on data center construction in the not-very-distant future.”* Efficiency gains in infrastructure will arrive, but Altman’s warning underscores the reality: the growth in usage is exponential, and infrastructure lags behind.

Adding to the pressure, GPT-5's enhanced reasoning capability carries a heavy energy toll. A study by the University of Rhode Island's AI lab found that a 1,000-token GPT-5 response consumes on average 18.35 Wh, compared to just 2.12 Wh for GPT-4—an increase of approximately 8.6× in energy per prompt. At ChatGPT-scale usage of around 2.5 billion requests per day, this translates to a staggering 45 GWh per day, equivalent to powering two to three nuclear reactors or about 1.5 million U.S. households. What seems like a modest per-prompt increase compounds massively when adopted globally. Eric Schmidt's insight is especially prescient here: *“Every doubling in capability triggers a compounding need for compute and power infrastructures.”* Scaling GPUs is tough, but scaling power grids, generation, transmission is a far longer-lead challenge. Efficiency improvements will come, but history shows they're invariably outpaced by adoption: smarter models drive heavier use, and heavier use accelerates the demand curve further. Add in that this race to tool launches is funded through cash flow and supported by the government. Historically, this was funded by debt or revenues.

The scale challenge becomes even more vivid when viewed through the lens of token throughput. On a Q1 earnings call, Microsoft CEO Satya Nadella revealed that the company processed *“over 100 trillion tokens this quarter, up 5× year-over-year, including a record 50 trillion tokens last month alone,”* underscoring that AI workloads are quickly becoming an everyday utility. At Google I/O, Sundar Pichai announced that monthly token processing had jumped from 9.7 trillion in April 2024 to 480 trillion in April 2025, a 50× increase, and that the number doubled again by their Q2 earnings call to nearly one quadrillion per month. Demis Hassabis later confirmed: *“We processed almost 1,000,000,000,000,000 tokens last month, more than double the amount from May.”* Tokens are not just abstract units of computation; every token generated consumes real compute cycles and therefore electricity. When growth is measured in the hundreds of trillions, the translation into energy demand is immediate and immense. Efficiency may lower the cost per token, but with throughput scaling this quickly, total compute and power needs are accelerating even faster than infrastructure can keep up.

The exponential demand doesn't end here with GPT-5. There will be new model releases from Google and xAI in the coming weeks which will continue pushing us closer to AI agents. Once agents are unleashed, they will increase energy demand significantly. These “insatiable digital shift-workers” never log off, never sleep, and continuously generate demand around the clock need to be fed and they eat power. Agents are where demand really escalates. AI agents, systems capable of planning, recalling, acting, and reasoning autonomously, can generate up to 25× more tokens per session through multi-step workflows, tool use, and API chaining. Adoption is fast: 79% of organizations are piloting or deploying agents, with 19% at scale. Their business impact is already tangible as 66% of adopters report productivity gains, and 88% plan to increase budgets to support agentic workflows. Yet agent workloads are among the least sustainable: dynamic, multi-turn reasoning consumes escalating infrastructure resources with diminishing efficiency returns. Here the efficiency paradox is stark: even if each agent call is made more efficient, the fact that agents run longer chains of calls means aggregate demand keeps accelerating. Efficiency buys time, but agents consume it.

Efficiency isn't the answer alone. The question isn't whether efficiency will improve, it always does. Anyone still on 2G with their smartphone? The question is whether it can advance fast enough to offset the compounded effects of capability growth and agentic proliferation. As Eric Schmidt has been saying in interviews lately on the topic, *“what Grove giveth, Gates taketh away”* using the Intel vs Microsoft famous supply and demand Silicon Valley line about efficiency. Every efficiency gain unlocks more demand. We've seen this before, from 2G to LTE and beyond, bandwidth usage exploded because richer modalities followed fast. In AI, efficiency paradoxes abound: lower cost per token simply enables more, richer, longer interactions, like video generation, which is far more compute-intensive than text. If video synthesis hit parity with human creativity, every film or advertisement could be generated in AI, a volume that efficiency improvements alone could not contain. Disney is estimated to have spent all in 1 billion dollars for Avengers: Endgame. What would their bid be on all the video capacity available?

And the stakes become even higher once AI leaves the screen and enters the physical world. Video generation may already push compute to its limits, but self-driving vehicles, and soon humanoid robots, require instantaneous perception and reasoning stacks with zero tolerance for error. As Andrej Karpathy, who served as the director of artificial intelligence and Autopilot Vision at Tesla, explained, *“vision is necessary in a sense that the world is designed for human visual consumption... and it is sufficient because it has all the information you need for driving. Humans obviously use vision to drive, so it's both necessary and sufficient.”* He further noted that humanoids must share this interface, since *“the world is designed for humanoid form factor... they would be able to operate our machines, sit down in chairs, potentially even drive cars.”* But processing vision in real-time with zero room for hallucinations is computationally heavy. Transforming brightness values into 3D, predicting motion paths, and planning actions, all under millisecond latency and power

constraints, requires massive offline training on supercomputers like Dojo to build accurate 3D reconstructions, paired with highly optimized chips inside cars or robots, where *“you have a finite budget of flops... and you must squeeze as much compute as you can into the tiny [car].”* Elon Musk has warned of the stakes plainly: if an AI system hallucinated one pedestrian, *“someone could die.”* Efficiency in this context doesn’t reduce the total demand, it only lowers latency or improves reliability. The physical world sets a hard floor on compute needs.

The strain is already showing up in the headlines. Utilities across the U.S. report being flooded with requests from data center developers, many tied to AI workloads, at volumes that dwarf today’s electricity use. As the Wall Street Journal described, *“a surge in speculative requests from data center developers for future AI needs has created a ‘phantom’ demand for electricity that, in some cases, exceeds the entire current consumption of utility service areas. This has left power companies in a difficult position, as they must balance the risk of building costly infrastructure for projects that may never materialize against the immense and rapidly growing power requirements of the AI industry.”*

Eric Schmidt emphasized this point in April at a Congressional hearing saying, *“We need energy in all forms, renewable, non-renewable, whatever. It needs to be there, and it needs to be there quickly.”* This statement can be supported by the news Microsoft reviving the Three Mile Island nuclear plant, Elon Musk importing an overseas power plant, a resurgence in coal, federal efforts to fast-track permitting through the AI Action Plan and on May 23, 2025 there was an executive order titled “Deploying Advanced Nuclear Reactor Technologies for National Security.” The news pattern is unmistakable: power is becoming the scarce resource, the gating factor in AI’s future. Chips can be built in cycles of months to a few years, but grids and gigawatts move on decade-long timelines.

So far in 2025, much of AI’s power demand has been met by tapping into excess capacity on existing grids and within data centers. But that runway is ending quickly. Think of a house wired with 100 wall outlets and only 20 appliances plugged in, each drawing modest electricity, that was AI in its early phase. And those extra 80 outlets were available only because, outside of AI, the broader economy has been sliding into recession, leaving industrial and commercial demand flat or falling. Today, nearly every outlet in the house is spoken for, and the new appliances aren’t toasters and lamps anymore, they’re energy-hungry dryers, ovens, and air conditioners that are about to run at high all day. We don’t just have a capacity issue in terms of outlets; we are approaching the point where the circuits themselves risk being overloaded. Meeting future AI demand will require not just plugging into what already exists, but building entirely new infrastructure, more circuits, bigger panels, and ultimately more power plants.

Until now, the AI infrastructure story has been told through NVIDIA and its chips, the GPU arms race that defined 2023 and 2024. But ultimately chips can scale far faster than power. New fabs are announced, new architectures are released, and chip performance doubles every one to two years. Nvidia has competition and their declining revenue growth shows it. Power, by contrast, is bound to the physics of grids, permitting processes, and energy infrastructure that take decades to build. This mismatch is why the bottleneck is shifting. Data centers can be filled with GPUs, but if the megawatts are not available, the chips sit idle. As GPT-5’s 8× energy intensity over GPT-4 makes clear, every new model compounds not just compute, but electricity needs. Efficiency gains will arrive but, as Schmidt warned, each gain is consumed immediately by new use cases. The real question now is whether the world can expand energy supply fast enough to sustain exponential demand. NVIDIA may remain the face of AI’s growth, but the defining story from here is power the gigawatts, grids, and generation required to keep the AI era alive. For investors, just remember, Nvidia’s market cap is 4 trillion dollars. The entire US “power” sector, energy, utilities and materials, have a combined market cap of 3 trillion dollars. Be ready for the next stage of AI demand driven story. Bob and weave to avoid that uppercut.

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